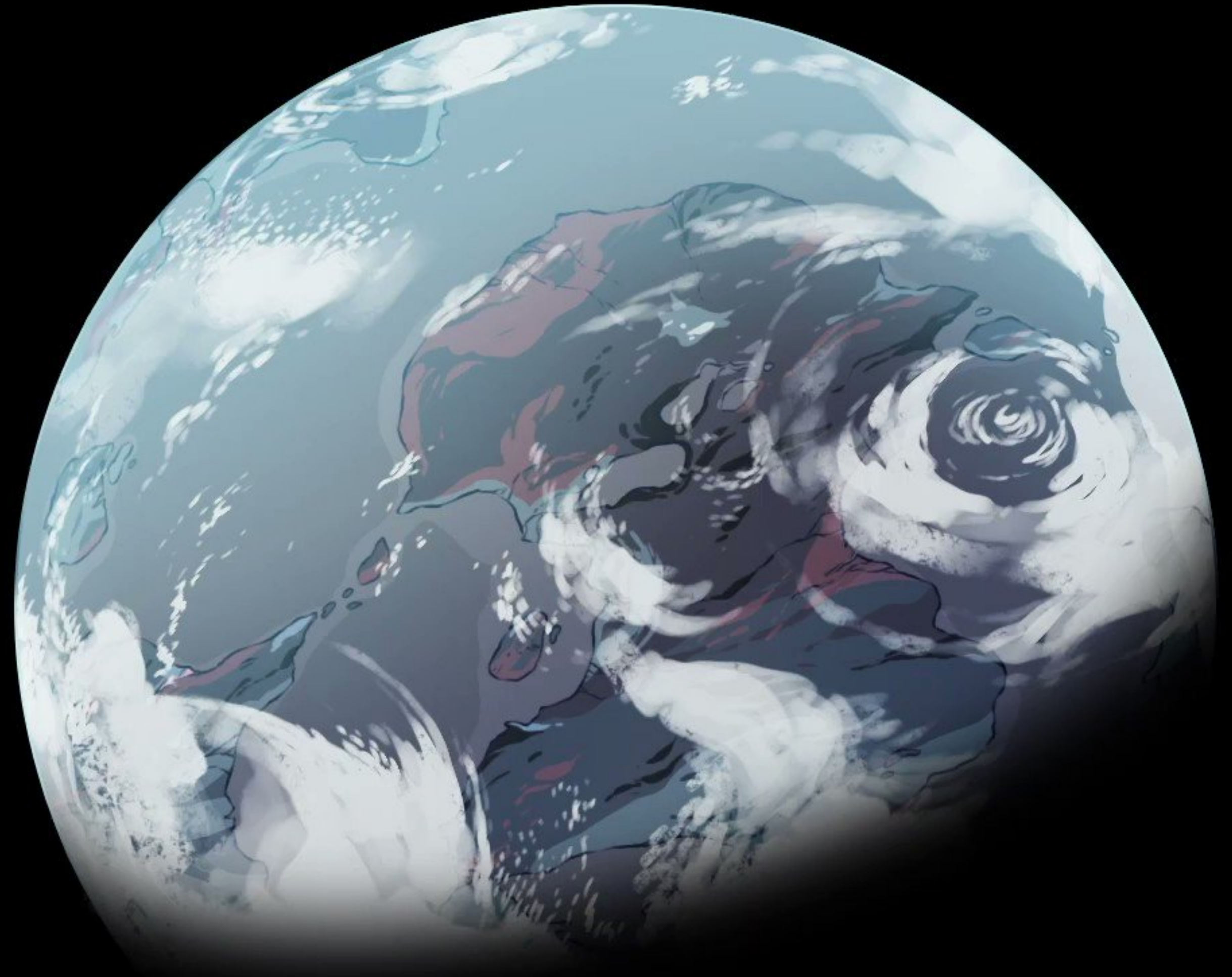


Science & Science Fiction

33-120 | Prof. Katie Breivik



Announcements

- Quiz 2 available now!
 - 85 pts — don't miss it!
 - Covers chapter 2: What is the nature of space and time?

bosons: fundamental *force* communicators

fundamental force

boson

electromagnetism

photon

weak nuclear force

(radioactive decay)

$W^{\pm}(\text{eak}), Z(\text{ero})$

strong nuclear force

(binds nucleons together)

gluon

gravity

graviton?

Einstein's field equations (1915-1920s)

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Curvature of
spacetime

Energy &
momentum

Einstein's field equations (1915-1920s)

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Matter tells spacetime
how to curve

Spacetime tells
matter how to move

Einstein's field equations (1915-1920s)

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Describes how objects moving influences spacetime and vice versa

*Predicts an unstable solution for
non-expanding or non-contracting spacetime*

Einstein's field equations (1915-1920s)

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

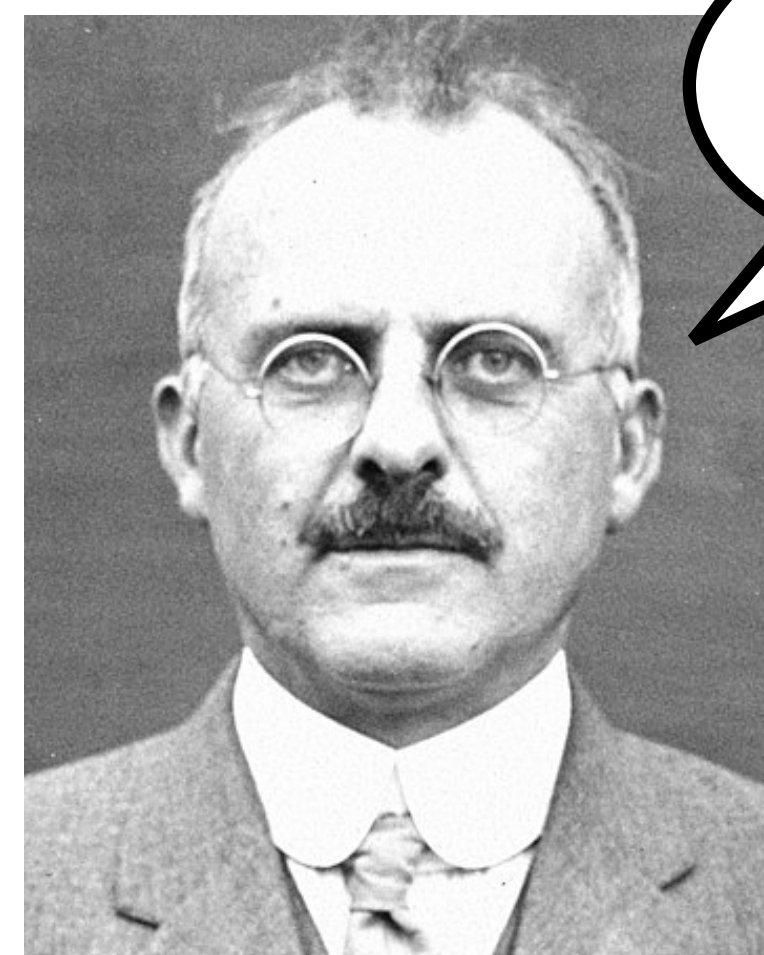
A constant
positive pressure
that balances out
the gravity

The Great Debate (1920)

Is the great spiral nebula just
dust in our Galaxy
or a separate galaxy?



It's dust!



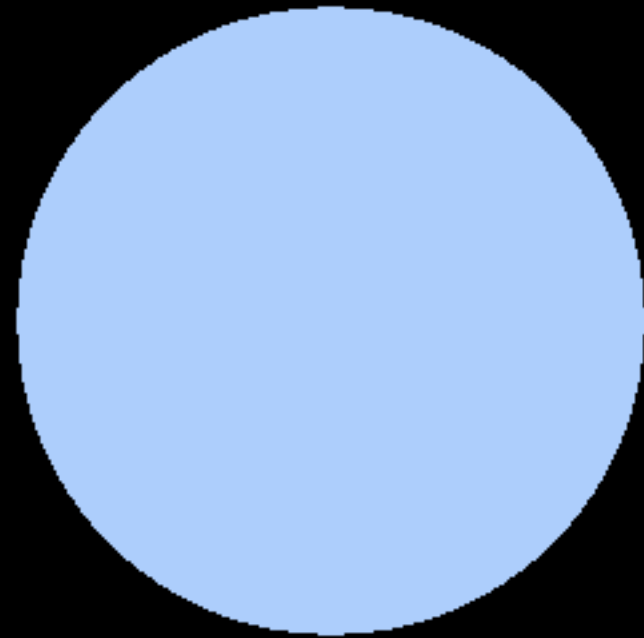
It's another
galaxy!!

Harlow Shapley

Heber Curtis



Cepheid variable stars



Pulse period $\longleftrightarrow$ absolute brightness

33-120: Science and Science Fiction



Henrietta Swan-Leavitt



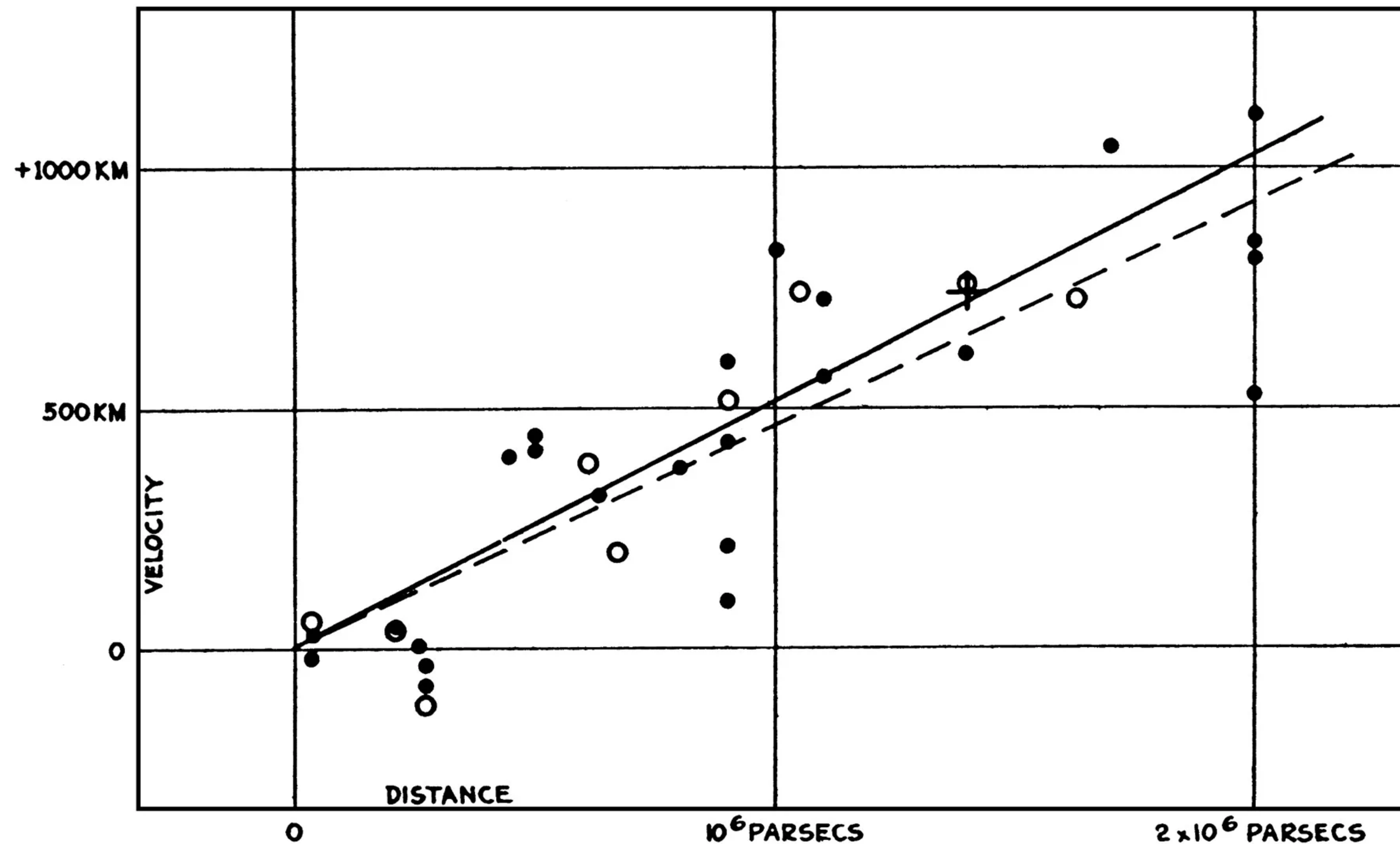
Edwin Hubble (1924)



Hooker 100in
telescope

Andromeda is *definitely* a different galaxy





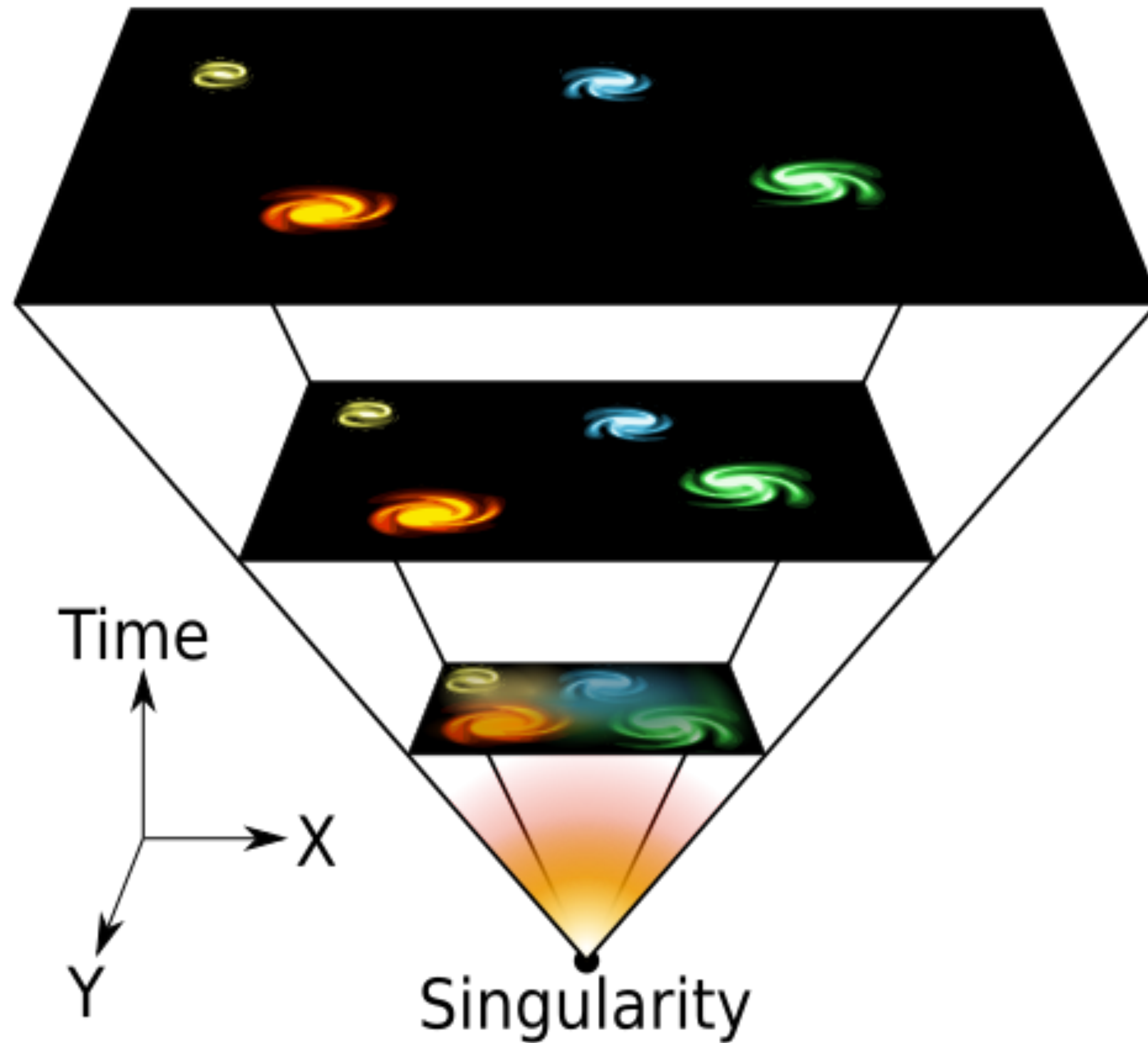
galaxies that
are farther
away are
moving away
from us
faster than
galaxies
close to us

The Universe is expanding!

Einstein's field equations (1930s-1990s)

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The Universe is expanding — the constant
was Einstein's “biggest blunder”



George LeMaitre

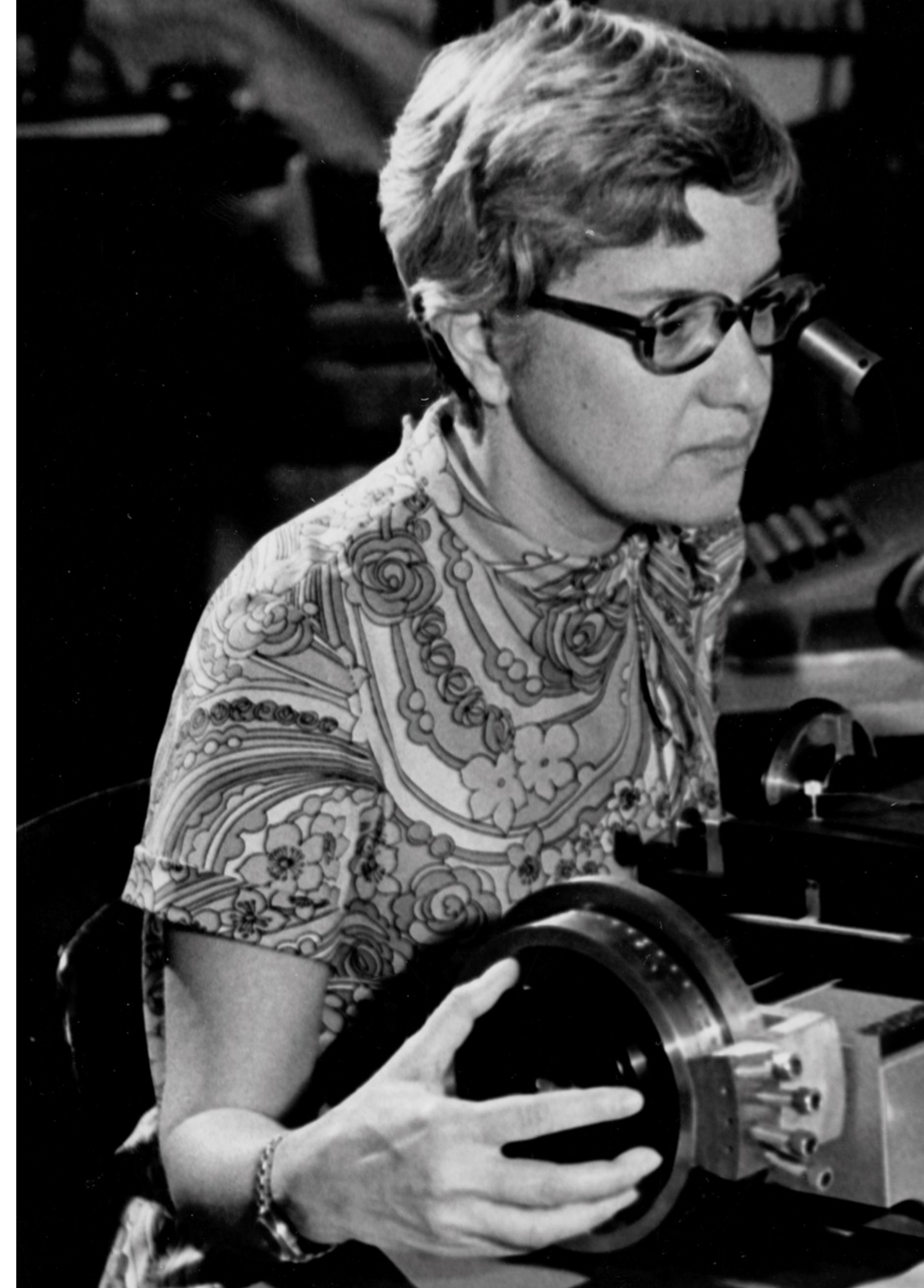
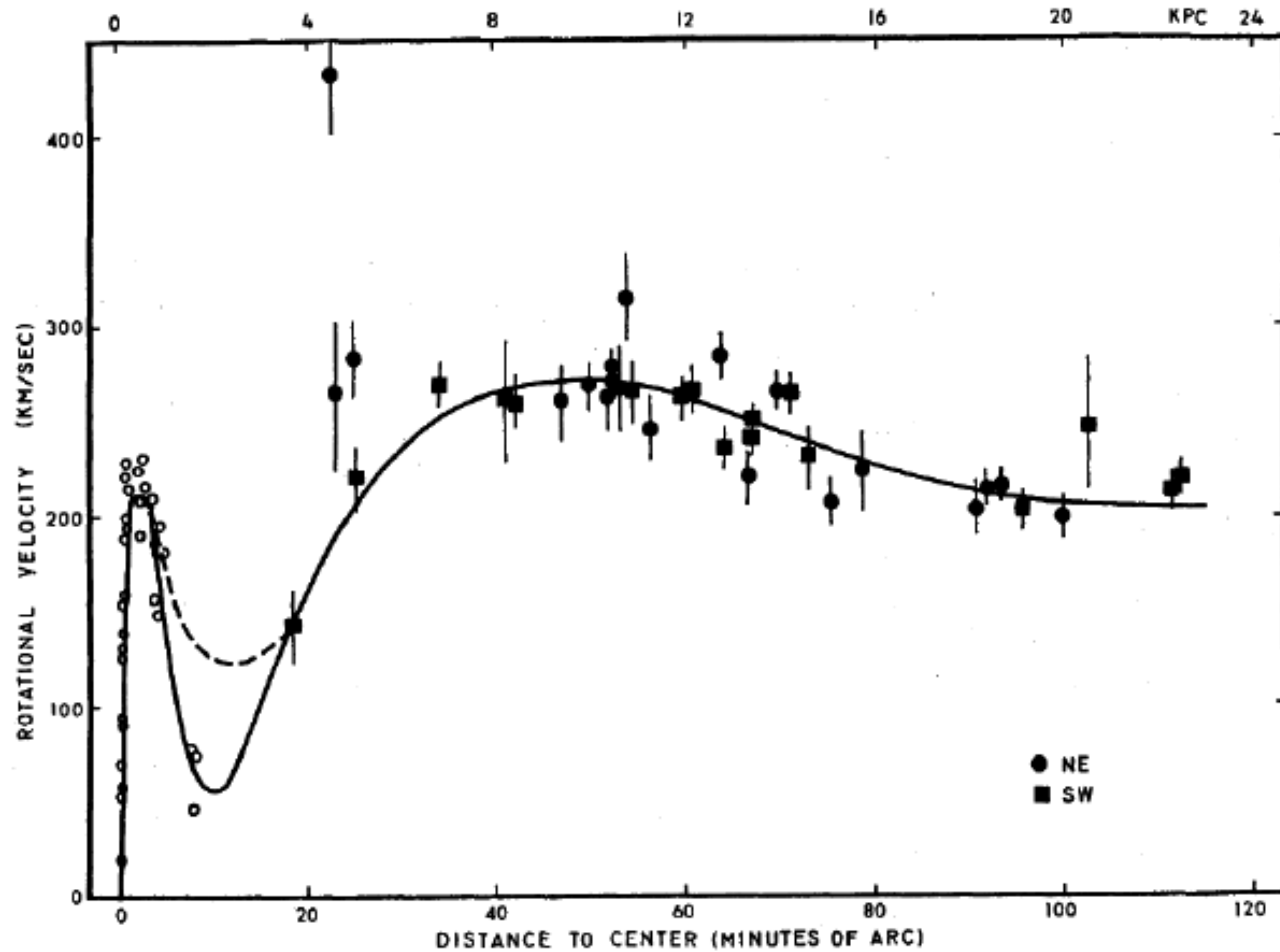


George Gamow

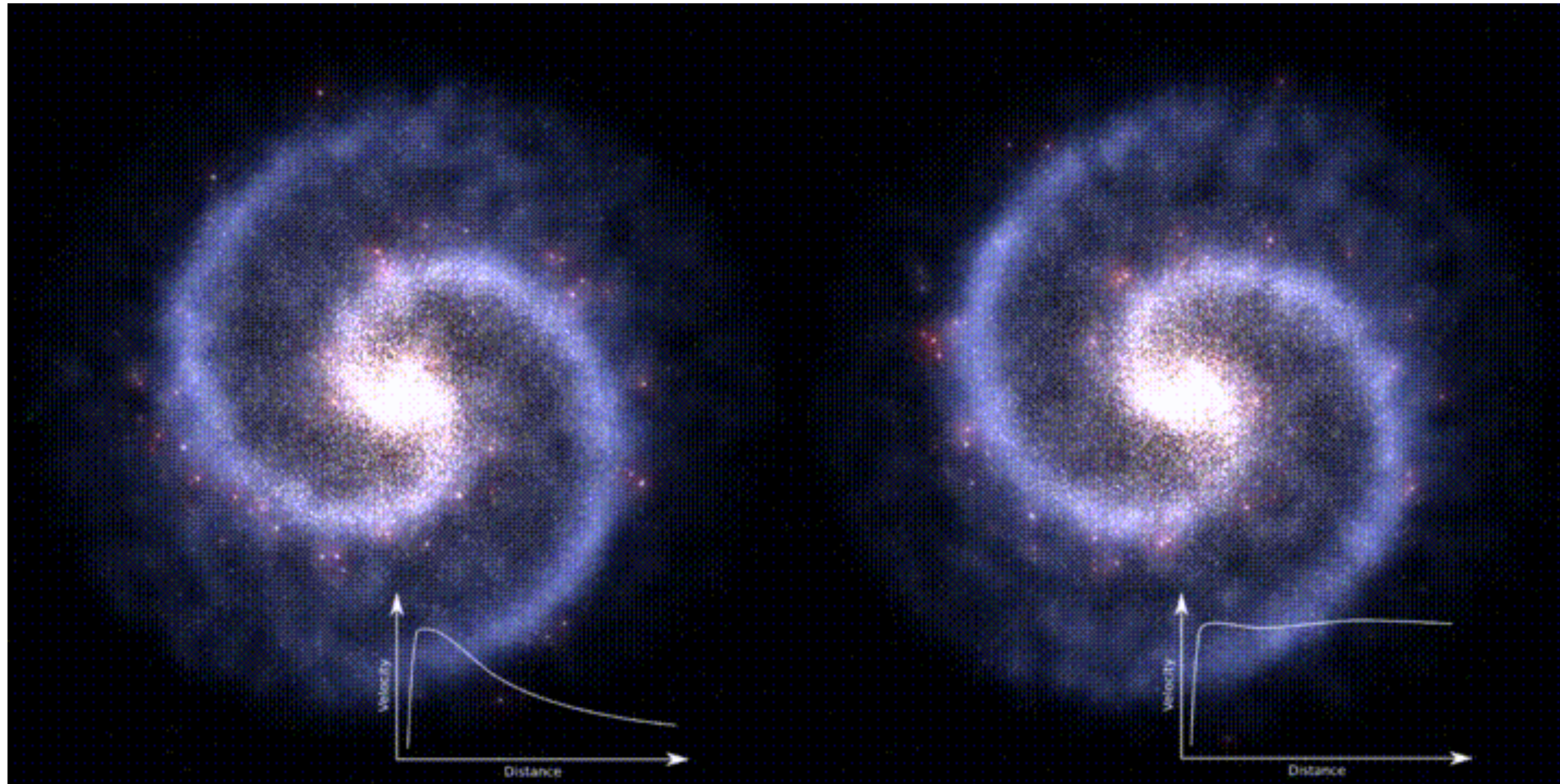
The Big Bang* (1931)

the first comprehensive
theory of cosmology

*named by Fred Hoyle who *didn't believe it*



Vera Rubin (1970)



What we expected
based on star light

What we see —
invisible matter!



Star Trek:
The Next Generation
“In Theory”

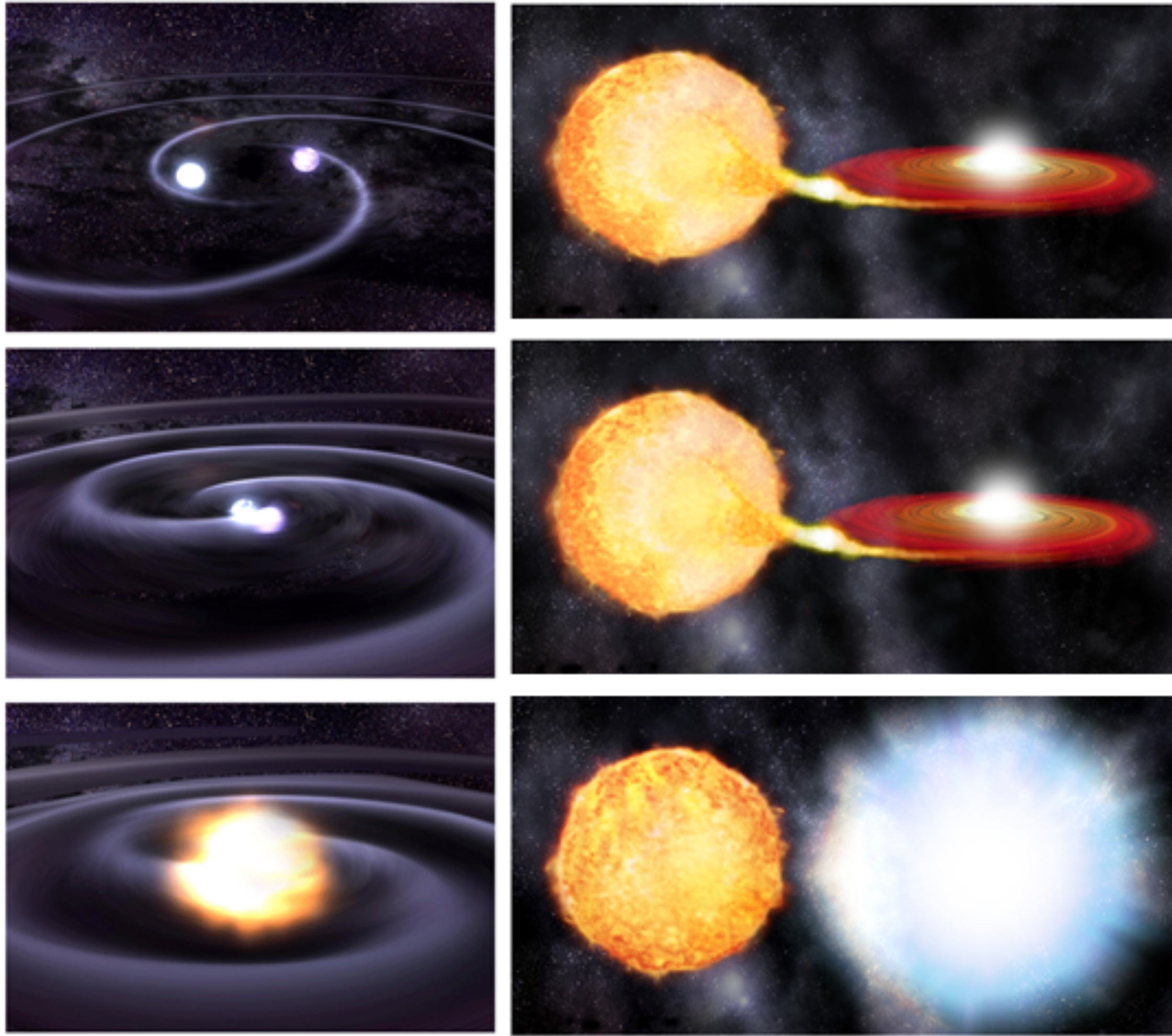
Written:
Gene Roddenberry

Paramount (1991)

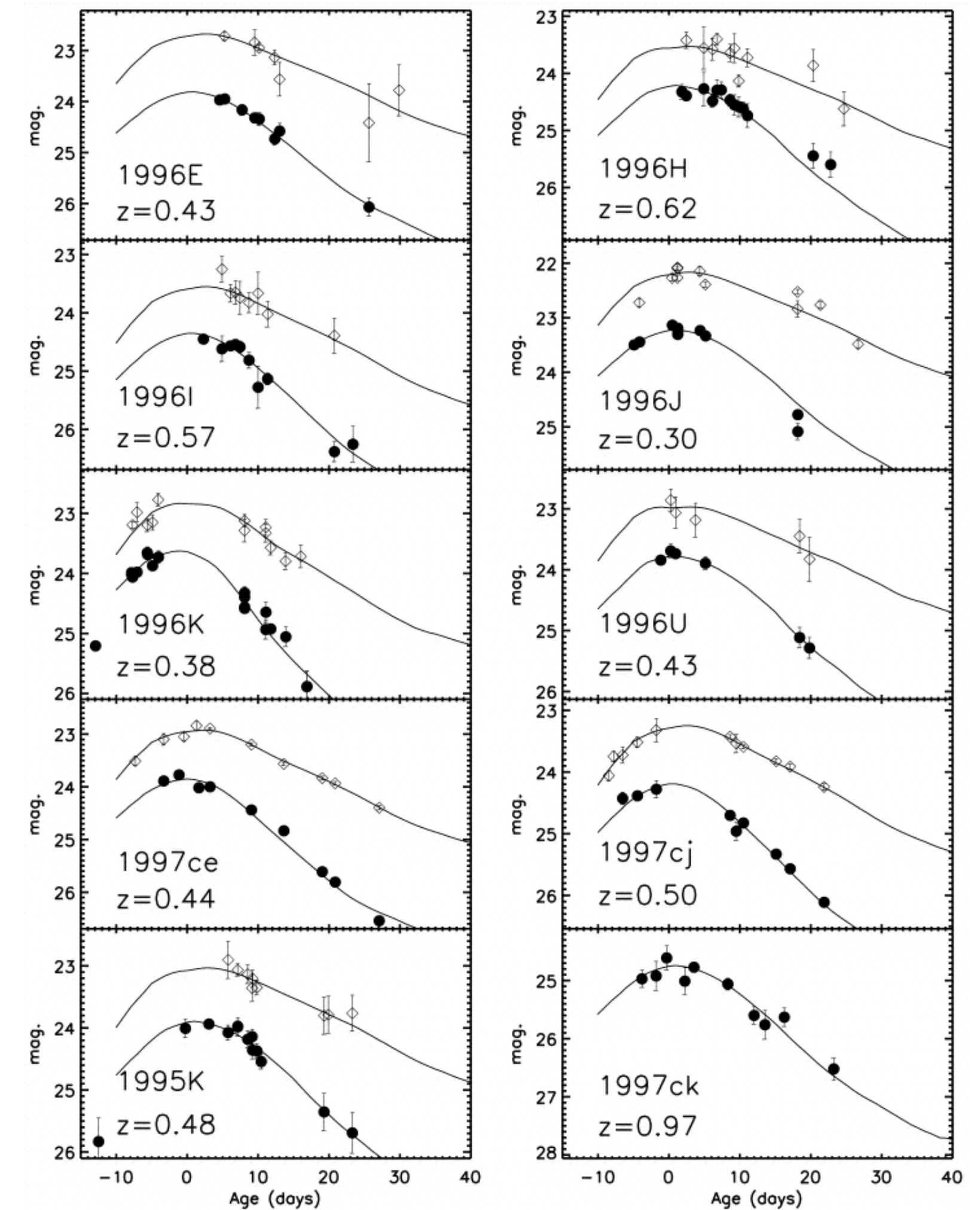
What do we know about dark matter?

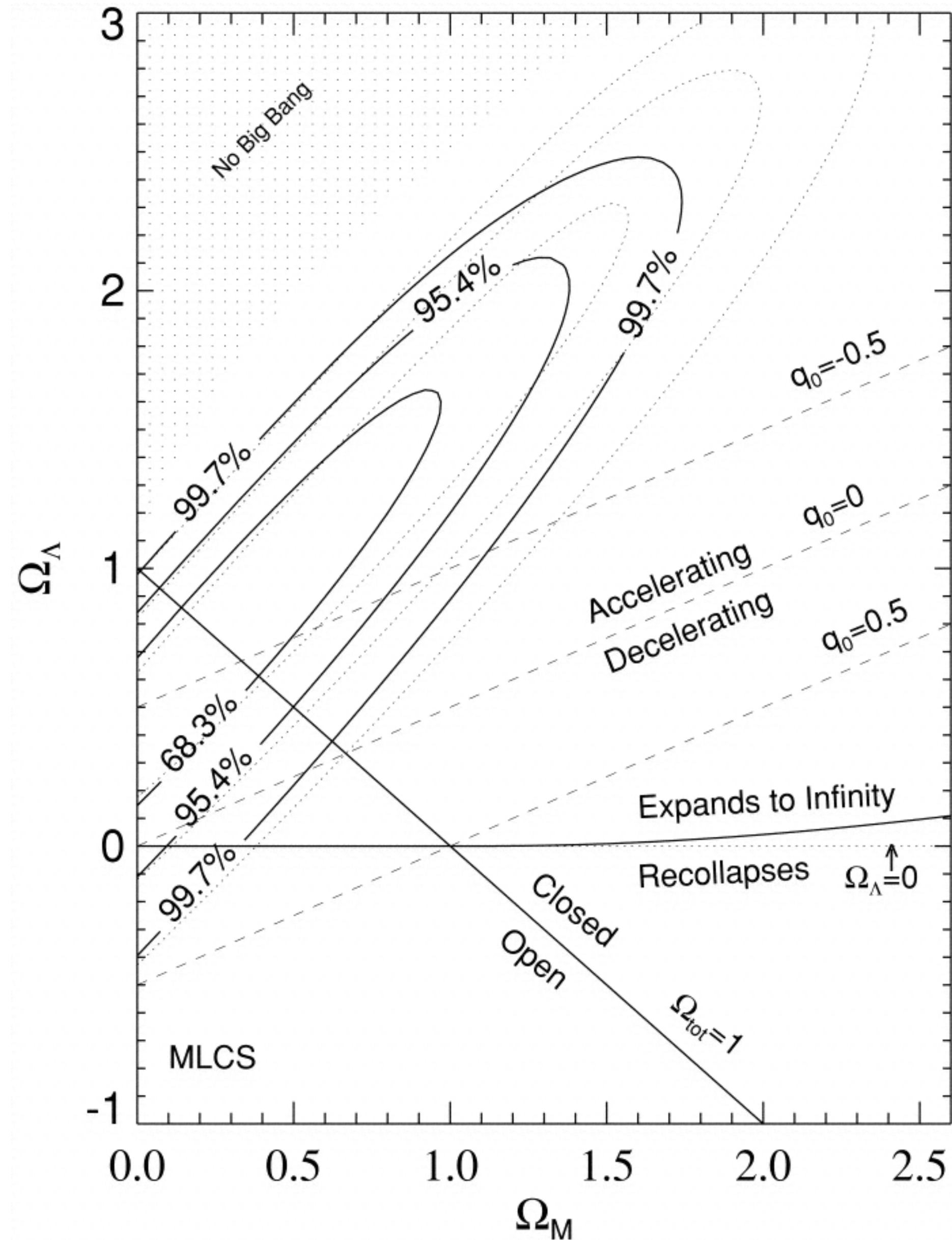
- ▶ There is a lot of it — 84% of the matter in the Universe!
- ▶ It is ubiquitous throughout the Universe — it's everywhere!
- ▶ It interacts *gravitationally* — makes galaxies rotate faster
- ▶ It does not interact *electromagnetically* — we can't see it

Type Ia supernovae: exploding white dwarfs



We know
how far
away
they are
because
of their
light
curve
shape





Adam
Riess



Saul
Perlmutter



Brian
Schmidt

The Universe is expanding at
an accelerated rate - 1998

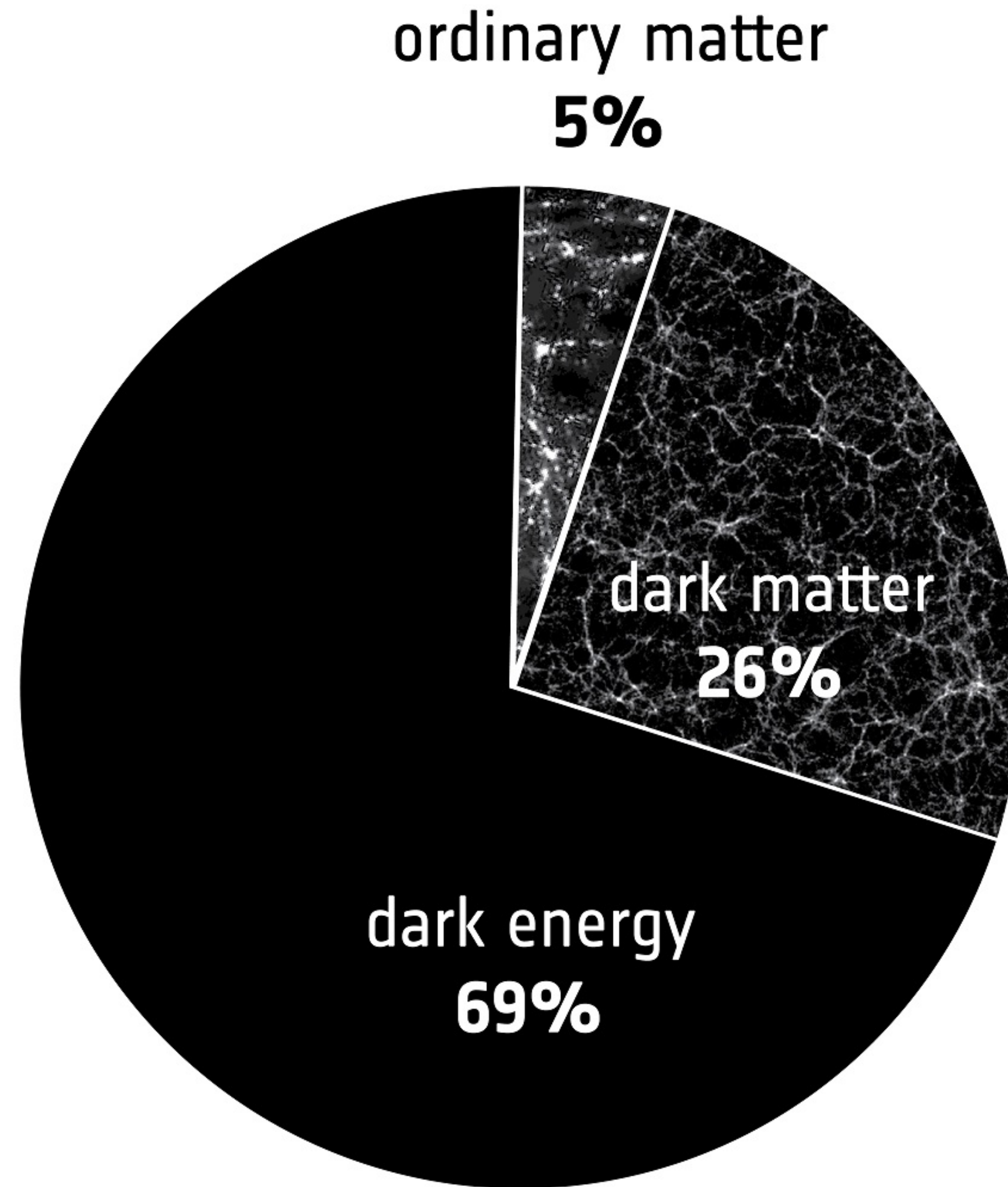
Einstein's field equations (1990s-today)

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Or maybe we don't get gravity ? Dark energy causes accelerated expansion Dark matter adds mass/momentum

The energy budget of the Universe

The standard model of particle physics only describes 5% of the energy in the Universe



Next time:

▶ Nuclear physics and radioactive decay

Reminders:

▶ Quiz 2 today!!